

Divy Patel

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Education

Arizona State University (ASU), Tempe, AZ

Aug. 2022 – Dec. 2023

Master of Science in Computer Science (MScS) (GPA: 3.55/4)

Pandit Deendayal Energy University (PDEU), India

Apr. 2018 – May 2022

B.Tech in Information and Communication Technology (GPA: 9.22/10)

Technical Knowledge

Programming Languages: Java, Python, R, C++, C#, JavaScript, SQL (MySQL, PostgreSQL), GraphQL, HTML

Frameworks: RESTful APIs (Microservices), Django, LangChain, TensorFlow, PyTorch, OpenCV, Scikit-learn

Tools and Technologies: AWS (Lambda, EC2, S3, DynamoDB), Docker, CI/CD, Git, Redis, Jira, Agile, Scrum

Work Experience

Walmart (Global Logic), Dallas, Texas

Software Engineer

Oct. 2024 – Present

- Developed Python/Django RESTful APIs enabling external TV control through mobile, HDMI-CEC, and IoT devices.
- Integrated new hardware abstraction layer into TV OS components, removing manufacturer library dependency across chipsets.
- Built a Generative AI-based BDD test generation pipeline for Behave feature files and step definitions, cutting AI cost per module from \$300-400 to \$40-60.
- Migrated smartcast-platform-layer TV middleware from Python to MicroPython via AI code transformation, validated by BDD suites.
- Automated defect tracking and reporting, saving the team lead over 6 hours weekly.
- Consolidated five middleware release branches into one stable branch, eliminating cherry-pick overhead and fix divergence.

Arizona State University, Tempe, Arizona

Engineer

Jul. 2024 – Dec. 2024

- Configured Linux servers and installed dependencies for the website, database, and ML needs.
- Assessed team requirements and recommended hardware for better performance and cost-efficiency.
- Contributed to the development of an LLM to generate daily tasks for artists based on their long-term goals.

Arizona State University, Glendale, Arizona

Research Technician

Apr. 2024 – Oct. 2024

- Developed programs to analyze genome data according to the lab's requirements under the supervision of Dr. Sreetharan Kanthaswamy.
- Contributed to a project analyzing genetic diversity and mutations across different monkey breeds.

OleOleSport, Remote (Santa Monica, CA)

Software Engineer

Apr. 2024 – Jul. 2024

- Built a football history chatbot using Langchain, Mistral 7B, and architectures like RAG and CRAG.
- Experimented with models, including Llama, achieving **98%** accuracy for Arsenal team history Q&A.
- Designed and developed a Flutter-based front-end for a football fan community website.

Quartic.ai, Remote

Data Engineer

Jul. 2021 – Jun. 2022

- Built real-time data pipelines pre-processing live data feeds with Python, GraphQL, and Redis, deployed via CI/CD.
- Optimized the data pipeline feeding ML models for predictive analytics and real-time risk monitoring.
- Developed a live risk monitoring dashboard on pipeline outputs using Grafana and ReactJS.

HOPS Healthcare, India

Artificial Intelligence Intern

Apr. 2021 – Dec. 2021

- Achieved **85%** accuracy in mobile-based skin disease detection by implementing OpenCV, TensorFlow and CNNs.
- Developed a noise reduction algorithm in Python, aiding detection of heart conditions from stethoscope audio.

Projects

LLM for Document Based Question Answering

Project

- Developed Large Language Model (LLM) using RAG implementation to interpret PDFs, CSV, and JSON files, with **98%** accuracy.
- Used LangChain for efficient storage and retrieval, ensuring accurate responses by implementing Prompt Engineering.
- Enhanced chat automation processes by testing model performance on diverse and complex queries.

Smart Parking Database Management System

PDEU

- Led development of comprehensive database management system for smart parking lot application.
- Generated 24,000 dummy data entries to thoroughly test and validate the database system.
- Documented SQL queries, query output snapshots, and CSV exports for database testing and analysis.

Graph Operations and Search

ASU

- Developed a Java application for graph operations with DOT parsing, node and edge management, and visualization using OOP patterns.
- Implemented Breadth-First Search (BFS) and Depth-First Search (DFS) with unit tests covering core operations.
- Provided a command-line interface for graph interaction, searches, and export to .dot and image formats.

Recent Achievements

● Publication in "Materials Today: Proceedings" on "[Artificial intelligence powered material search engine](#)"

Apr. 2022

● Publication in American Journal of Primatology on "[MHC Diversity in Captive Rhesus Macaque Colonies](#)"

Jan. 2026